

Presentation Transcript

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June 5, 2026

The classic Brunn-Minkowski Inequality on $\mathbb{R}$ is the following. In this very simple case the inequality is provable directly with elementary techniques.

BRUNN-MINKOWSKI INEQUALITY ON $\mathbb{R}$

Let $\lambda \in (0, 1)$ and $X, Y \subseteq \mathbb{R}$ be bounded measurable sets s.t. $(\lambda - 1)X + \lambda Y$ is measurable. Then,

$$\text{vol}((1 - \lambda)X + \lambda Y) \geq (1 - \lambda) \text{vol}(X) + \lambda \text{vol}(Y)$$

Proof.

By inner regularity of the Lebesgue measure it suffices to consider X, Y compact. We first prove that $X + Y$ is compact. Let $\{x_n + y_n\}_{n \in \mathbb{N}}$ be a sequence of $X + Y$ and we consider $\{x_n\}_{n \in \mathbb{N}} \subseteq X$ and $\{y_n\}_{n \in \mathbb{N}} \subseteq Y$. By compactness of X there exists a convergent subsequence $\{x_n\}_{n \in A} \subseteq \{x_n\}_{n \in \mathbb{N}} \rightarrow x \in X$, and then by compactness of Y there exists a convergent subsequence $\{y_n\}_{n \in B} \subseteq \{y_n\}_{n \in A} \rightarrow y \in Y$. Hence $\{x_n + y_n\}_{n \in B} \rightarrow x + y \in X + Y$ is a convergent subsequence.

By compactness there exists $x_0, y_0 \in \mathbb{R}$ s.t. $X + \{x_0\} \subseteq \mathbb{R}^-$ and $Y + \{y_0\} \subseteq \mathbb{R}^+$ with $(X + \{x_0\}) \cap (Y + \{y_0\}) = \{0\}$. Using the origin we have $X + Y + \{x_0 + y_0\} \supseteq (X + \{x_0\}) \cup (Y + \{y_0\})$. Thus,

$$\text{vol}(X + Y) = \text{vol}(X + Y + \{x_0 + y_0\}) \geq \text{vol}(X + \{x_0\}) + \text{vol}(Y + \{y_0\}) = \text{vol}(X) + \text{vol}(Y)$$

And the proposition follows from letting $X \rightarrow (1 - \lambda)X, Y \rightarrow \lambda Y$ and positive-homogeneity of measure. ■

We may extend this result easily to finitely many sets as follows,

BRUNN-MINKOWSKI INEQUALITY FOR FINITELY-MANY SETS ON $\mathbb{R}$

Let $\lambda_1 + \dots + \lambda_n = 1$ and $X_1, \dots, X_n$ be bounded measurable subsets s.t. $X_1 + \dots + X_n$ is measurable. Then,

$$\text{vol}(\lambda_1 X_1 + \dots + \lambda_n X_n) \geq \lambda_1 \text{vol} X_1 + \dots + \lambda_n \text{vol} X_n$$

Proof.

This follows by induction on n . The base case $n = 2$ is given by the classic inequality. Assume up to $n - 1$ and,

$$\begin{aligned} \text{vol}(\lambda_1 X_1 + \dots + \lambda_{n-1} X_{n-1} + \lambda_n X_n) &= \text{vol}\left((1 - \lambda_n) \left(\frac{\lambda_1}{1 - \lambda_n} X_1 + \dots + \frac{\lambda_{n-1}}{1 - \lambda_n} X_{n-1} \right) + \lambda_n X_n\right) \\ &\geq (1 - \lambda_n) \text{vol}\left(\frac{\lambda_1}{1 - \lambda_n} X_1 + \dots + \frac{\lambda_{n-1}}{1 - \lambda_n} X_{n-1} \right) + \lambda_n \text{vol} X_n \\ &\geq \lambda_1 \text{vol} X_1 + \dots + \lambda_{n-1} \text{vol} X_{n-1} + \lambda_n \text{vol} X_n \end{aligned}$$

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One may consider extending this inequality to *countably-many* sets. However to do so induction is no longer sufficient as the procedure does not terminate. Still we have an intermediary result,

COUNTABLE SUM OF COMPACT SETS IS COMPACT

Let each element of $\{X_i\}_{i=1}^\infty$ be compact. Then if $\sum_{i=1}^\infty X_i$ is bounded then it is compact.

Proof.

Let $\{\sum_{i=1}^\infty x_{i,n}\}_{n \in \mathbb{N}}$ be a sequence of $\sum_{i=1}^\infty X_i$. By compactness of X_1 there exists a convergent subsequence $\{x_{1,n}\}_{n \in N_1} \rightarrow x_1 \in X_1$ where $N_1 \subseteq \mathbb{N}$. By compactness of X_2 there exists a convergent subsequence $\{x_{2,n}\}_{n \in N_2} \rightarrow x_2 \in X_2$ where $N_2 \subseteq N_1$. This procedure continues inductively yielding infinitely descending subsets $\cdots \subseteq N_2 \subseteq N_1 \subseteq \mathbb{N}$ s.t. the subsequence $\{x_{i,n}\}_{n \in N_i} \rightarrow x_i \in X_i$ converges. Define N to contain the i th element of N_i , i.e. the diagonal. Then $\sum_{i=1}^\infty \{x_{i,n}\}_{n \in N} \rightarrow \sum_{i=1}^\infty x_i \in \sum_{i=1}^\infty X_i$ is a convergent subsequence.

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Alternatively we may prove the Brunn-Minkowski Inequality using its analytic counterpart in the Prekopa-Leindler Inequality, which we suitably generalize to finitely-many functions as well. We consider the proof without using level sets, since that proof requires Brunn-Minkowski on $\mathbb{R}$, which we want to generalize.

PREKOPA-LEINDLER INEQUALITY ON $\mathbb{R}^n$

Where $\sum_i \lambda_i = 1$ and $h, f_i: \mathbb{R}^n \rightarrow \mathbb{R}$ are nonnegative measurable functions satisfying,

$$h\left(\sum_i \lambda_i \mathbf{x}_i\right) \geq \prod_i f_i(\mathbf{x}_i)^{\lambda_i}$$

For all $\mathbf{x} \in \mathbb{R}^n$. Then,

$$\int_{\mathbb{R}^n} h \geq \prod_i \left(\int_{\mathbb{R}^n} f_i \right)^{\lambda_i}$$

Proof.

We prove this by induction on n . For the base case $n = 1$ let $F_i = \int_{\mathbb{R}} f_i$ and it be positive or else the inequality is trivial. Consider the cumulative density function,

$$u_i: [0, 1] \rightarrow \mathbb{R}, \quad t \mapsto \inf_{t' \in \mathbb{R}} \left\{ \frac{1}{F_i} \int_{-\infty}^{t'} f_i = t \right\}$$

Now u_i may be discontinuous but is increasing so is differentiable a.e.. Define,

$$w(t) = \sum_i \lambda_i u_i(t)$$

And so is increasing being the sum of increasing functions. By differentiating the equality satisfied by u_i we obtain,

$$\frac{1}{F_i} f_i(u_i) \dot{u}_i = 1$$

Hence in conjunction with the AM-GM inequality,

$$\dot{w} = \sum_i \lambda_i \dot{u}_i \geq \prod_i \dot{u}_i^{\lambda_i} = \prod_i \left(\frac{F_i}{f_i(u_i)} \right)^{\lambda_i}$$

We now evaluate the integral of h using the change-of-variables $x = w(t)$,

$$\begin{aligned} \int_{\mathbb{R}} h &= \int_0^1 (h \circ w) \dot{w} \\ &= \int_0^1 h \left(\sum_i \lambda_i u_i \right) \sum_i \lambda_i \dot{u}_i \\ &\geq \int_0^1 \prod_i f_i(u_i)^{\lambda_i} \prod_i \left(\frac{F_i}{f_i(u_i)} \right)^{\lambda_i} \\ &= \prod_i \left(\int_{\mathbb{R}} f_i \right)^{\lambda_i} \end{aligned}$$

Therefore assume true up to $n - 1$. Define,

$$h^{(s)}: \mathbb{R}^{n-1} \rightarrow \mathbb{R}, \quad \mathbf{y} \mapsto h(\mathbf{y}, s), \quad f_i^{(s)}: \mathbb{R}^{n-1} \rightarrow \mathbb{R}, \quad \mathbf{y} \mapsto f_i(\mathbf{y}, s)$$

Observe for a fixed s that,

$$\begin{aligned} h^{(\sum_i \lambda_i s_i)} \left(\sum_i \lambda_i \mathbf{y}_i \right) &= h \left(\sum_i \lambda_i \mathbf{y}_i, \sum_i \lambda_i s_i \right) \\ &= h \left(\sum_i \lambda_i (\mathbf{y}_i, s_i) \right) \\ &\geq \prod_i f_i(\mathbf{y}_i, s_i)^{\lambda_i} \\ &= \prod_i f_i^{(s_i)}(\mathbf{y}_i)^{\lambda_i} \end{aligned}$$

So in particular obeys the assumption of the inductive hypothesis. Therefore we may conclude,

$$H \left(\sum_i \lambda_i s_i \right) := \int_{\mathbb{R}^{n-1}} h^{(\sum_i \lambda_i s_i)} \geq \prod_i \left(\int_{\mathbb{R}^{n-1}} f_i^{(s_i)} \right)^{\lambda_i} =: \prod_i F_i(s_i)^{\lambda_i}$$

Namely if we define H to be the LHS and $\prod_i F_i$ to be the RHS, and these to be functions of the superscripts, then they too obey the inductive hypothesis on $\mathbb{R}$. Written out we conclude,

$$\int_{\mathbb{R}^n} h = \int_{\mathbb{R}} H \geq \prod_i \left(\int_{\mathbb{R}} F_i \right)^{\lambda_i} = \prod_i \left(\int_{\mathbb{R}^n} f_i \right)^{\lambda_i}$$

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Largely speaking the parts following the inductive hypothesis is simply applying the one or $(n - 1)$ -dimensional case. Therefore to generalize the Prekopa-Leinder Inequality to countably-many functions is to generalize the one-dimensional case. In particular we see that this is largely a matter of determining under what conditions we have the convergence of the function $w(t = \sum_i \lambda_i u_i(t))$, i.e. the cumulative density function, and when we may commute the infinite sum and the derivative in order to obtain the same AM-GM inequality.